CS771 (Autumn 2026)
Mini-Project 1

Next-Day PM2.5 Forecasting and Heatmap Generation for Bihar

with cleaning of a raw Kanpur sensor dataset

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Report covering Parts 0–5 · 30 October 2026

Summary

All errors are in µg/m^3. Parts 1–5 numbers are from `IML Part 1 to 5.ipynb` and its `outputs/` files; Part 0 numbers are computed from our cleaned file `kanpur_hourly_clean.csv` and the raw Kanpur file. Validation = every station and every 12:00 target from 1 Sep to 30 Nov 2024 (91 days × 468 stations = 42,588 forecasts). Only scikit-learn models from Lectures 1–16; no neural networks, no external data.

Day-ahead forecast	RMSE	MAE	RMSE Sep / Oct / Nov	MAE Sep / Oct / Nov
Persistence (tomorrow = today)	32.75	17.47	23.33 / 20.22 / 47.81	8.91 / 11.97 / 31.70
Part 1 Ridge (29 raw features, $\alpha = 1$)	29.58	18.16	20.34 / 19.36 / 43.06	10.35 / 13.46 / 30.81
Part 2 Ridge (39 selected features)	29.08	16.57	19.49 / 17.92 / 43.05	8.07 / 11.51 / 30.31
Final: $\frac{1}{2}$ Ridge + $\frac{1}{2}$ Extra-trees	**28.77**	**16.13**	**19.26 / 17.78 / 42.58**	**7.72 / 11.30 / 29.53**

The final model beats both baselines on both metrics overall and in every month (by 3.97 RMSE / 1.34 MAE against persistence and 0.81 / 2.03 against the Part 1 Ridge). Its daily RMSE is lower than persistence on 73/91 days and lower than the Part 1 Ridge on 60/91 days (sign test $p = 2.5 \times 10^{-9}$ and $p = 0.0016$).

1 Part 0 — Cleaning the raw Kanpur data

Format

- 76 sensors, 1 Jul–31 Aug 2026 on a 15-minute UTC grid (5,952 slots = 1,488 hours): 44 sensors with 10-digit IDs, 31 with 8-character IDs and one 12-character ID (`AC0BFBE34CC7`).
- *Reporting:* 75 sensors report only at minute :00 (so 3 of every 4 slots are empty by design); only `AC0BFBE34CC7` reports every 15 minutes. Sensors report in 51–88 % of the 1,488 hours (median 74 %; median 1,184 hours for 10-digit, 1,014 for 8-char, 1,212 for the 12-char sensor).
- *No sensor moves:* each has a single latitude/longitude pair.
- *ID styles differ in resolution and level:* 10-digit sensors report whole numbers (median 37 µg/m^3), 8-char sensors report long decimals (99.8 % non-integer; median 33 µg/m^3), the 12-char sensor whole numbers (median 24.5 µg/m^3).
- *Hourly long table* (`sensor_id, timestamp_utc, latitude, longitude, pm25`): readings averaged within the hour, labelled by the hour start. This only changes the 15-minute sensor: its hourly mean differs from the :00 reading by 2.6 µg/m^3 on average (median 8 % of the value; 1,208 hours).

Problems detected and flagged (never deleted, gaps never filled)

Flag	Rule and justification of the threshold	Readings	Sensors
error code	value 999: it repeats 136 times, while no other value above 400 occurs more than once; the 99.9th percentile of all other values is 294	136	5
zero	value 0: in those hours the rest of the network reads a median of 35.9 µg/m^3, so a zero is a dead channel, not clean air	813	21
stuck	same value for ≥24 consecutive reported hours: runs of equal values fall from 219 (≥4 h) to 38 (≥12 h) and 11 (≥24 h), then level off (10 at ≥36 h and ≥48 h) — only genuine faults survive 24 h	929	7
dropped sensor	>50 % of a sensor's readings flagged: `1203260068` (58 %) and `1203260082` (73 %); the next worst sensor has 15 %, so the threshold is not critical	600	2
outage hour	fewer than 10 % of sensors report: one window of 117 hours (5 Aug 08:00 – 10 Aug 04:00 UTC); the count is the same for any threshold from 5 to 10 % (column `outage_hour`)	117 h	—

- *Sensors caught.* error code: `1204260025` (120 of 136), `1203260086`, `1203260075`, `1203260125`, `1203260084`. zero: mostly `1203260082` (414) and `1203260068` (350). stuck: `1203260082`, `1203260068`, `1204260005`, `1203260146`, `RZGVBSNC`, `ZQQZB912`, `1203260084`.
- *Weak followers (identified, not flagged):* correlation of each sensor's `ok` readings with the leave-one-out network median, threshold median − 2 sd = 0.69 − 2 × 0.16 = 0.37: `1204260025` (0.17), `U8APQDBX` (0.26), `1203260075` (0.29), `TUOBZVJ0` (0.36). Two of them also produce error codes.

- *Clean file* `kanpur_hourly_clean.csv`: 113,088 rows (76 sensors × 1,488 hours) with columns `pm25`, `flag`, `n_raw`, `outage_hour`; missing hours stay as rows with `flag = missing` and empty `pm25`.

Before → after	Sensors	Readings	Share of sensor-hours missing
Before (raw, hourly)	76	82,131 reported	27.4 %
After (`ok` only)	74	79,653 `ok` (97.0 % of reported)	29.6 % (missing or flagged)

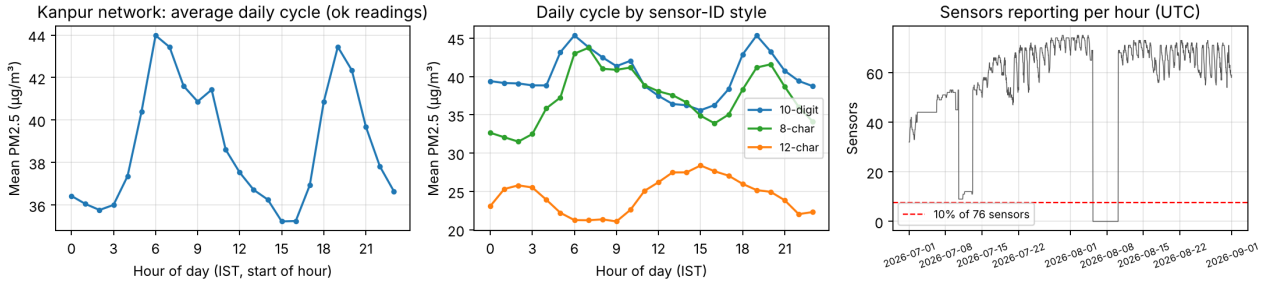


Figure 1: Left: Kanpur network daily cycle in IST (`ok` readings): peaks at 06:00 (44.0 µg/m^3) and 19:00 (43.4), minimum at 15:00–16:00 (35.2). Middle: the same by ID style; the 12-character sensor peaks in the afternoon, the opposite phase to the rest, which suggests its timestamps are local time labelled as UTC (a limitation we note; its values are not used in the calibration). Right: sensors reporting per hour; the 5–10 Aug outage is the only window below 10 %.

Calibration check (fit on July, test on August)

Each 10-digit sensor was paired with its nearest 8-char sensor within 3 km (assignment km formula): 23 pairs (median distance 2.4 km), 21 with overlapping `ok` readings (7,804 paired hours in July, 11,155 in August). A line fitted on July gives $\textit{8-char} \approx 29.7 + 0.21 \times \textit{10-digit}$.

- **August disagreement:** no correction RMSE 23.34 / MAE 12.33; with the July line RMSE 23.00 / MAE 15.27; a constant July offset (+1.11) gives 23.29 / 12.42. The correction barely moves RMSE and makes MAE clearly *worse*, so we do not apply it.
- **Why:** the systematic difference is small (August mean 8-char − 10-digit = +1.7 µg/m^3 at a level of ≈37–39) compared with the random disagreement (partner correlation only 0.49 in August). With such a noisy input, least squares learns a slope far below 1 (0.21, regression dilution): it pulls every forecast towards the July mean, which fits July (RMSE 23.68 → 17.09) but does not transfer. The two device families agree on average; what separates paired sensors is local noise and spikes, which a linear calibration cannot fix.

2 Part 1 — Required baselines

- One row = (station, 12:00 issue time); target = same station 24 h later. Train: targets 3 Jan–31 Aug (242 days, 113,256 rows; 1–2 Jan lack 24 h of history). Validation: 42,588 rows (14,040 / 14,508 / 14,040 in Sep / Oct / Nov), nothing dropped, no shuffling.
- Ridge(alpha=1) on the 29 raw features (24 hourly lags; mean of the 10 nearest other stations now and over 24 h; network mean now; $\sin/\cos(2\pi d/365)$), pooled over all stations, negatives clipped to 0. Our run reproduces the reference numbers (table in the Summary).

Q: Does one baseline win on both metrics? Why can a squared-error model lose on absolute error? No. Ridge wins RMSE (29.58 vs 32.75, and in every month) but loses MAE (18.16 vs 17.47; it loses MAE in Sep and Oct and wins only in November, 30.81 vs 31.70). Squared error is minimised by the conditional *mean*, absolute error by the *median*; PM2.5 is right-skewed, so the mean lies above the typical value. Ridge also shrinks towards the training average, which contains the polluted winter: its mean bias is $+5.55/+6.86/+3.49$ in Sep / Oct / Nov, against $-0.04/-0.79/-0.58$ for persistence. This upward bias reduces the few very large errors that dominate RMSE but adds a small error to the many ordinary clean days, which is what MAE counts.

3 Part 2 — Improving the forecast

Feature groups (8 groups, 57 features)

All features use only readings at or before the issue time; the only statistic fitted outside the model (station mean) uses training rows only. Training rows for all Part 2 experiments: targets 9 Jan–31 Aug (236 days × 468 = **110,448 rows**, every station, equal weights), because the longest windows need 168 h of history. Validation is unchanged.

Group	n	Idea
`lags24`	24	station's readings at $t, t-1\text{h}, \ldots, t-23\text{h}$
`history_days`	10	same hour on each of the previous 6 days; 72 h and 168 h mean; 168 h std and min
`neigh_knn`	3	mean of the 10 nearest stations now, over 24 h and over 168 h
`neigh_radius`	6	mean of stations within 15 / 30 / 60 km, now and over 24 h
`network`	4	mean of all 468 stations now, over 6 h, 24 h and 168 h
`trend`	3	today − yesterday (same hour; 24 h mean; network 24 h mean)
`seasonal`	5	station training mean, 30-day mean, 30-day mean relative to network, sin/cos of day of year
`dow`	2	sin/cos of day of week

Group selection, models and ablation

Forward selection (Ridge, $\alpha = 1$; add the group that lowers validation RMSE most, stop when the gain is < 0.02): `lags24` 29.863 → `+network` 29.261 → `+neigh_radius` 29.176 → `+trend` 29.121 → `+dow` 29.078 (39 features). Adding the remaining groups made it worse or did not help: `+neigh_knn` 29.113, `+history_days` 29.330, `+seasonal` 29.620; all 8 groups 30.246.

Ablation of the selected Ridge (Δ = RMSE without the group − 29.078; positive = helps)

Removed group	n	RMSE without	ΔRMSE
`lags24`	24	36.111	+7.033
`network`	4	29.286	+0.209
`neigh_radius`	6	29.187	+0.109
`trend`	3	29.134	+0.057
`dow`	2	29.121	+0.044

Models on the same 39 features

Model	train RMSE	val RMSE	val MAE
Ridge ($\alpha = 1$)	26.92	29.08	16.57
Hist. gradient boosting	19.74	33.23	18.87
Random forest	19.07	30.71	17.41
Extra-trees	20.22	30.53	17.19
Mean(Ridge, Extra-trees)	—	29.44	16.58

Overlap. Every kept group earns its place, but all except `lags24` cost little when removed alone, because they carry similar information: on validation rows lag0h correlates 0.76–0.78 with the 15 km neighbour mean and the network mean, and the network mean now and its 24 h mean correlate 0.96 with each other and 0.92–0.96 with the 60 km neighbour mean.

Final model: $\frac{1}{2}$ Ridge + $\frac{1}{2}$ Extra-trees on the change from today

Both halves predict $\Delta = y - \text{today's reading}$ (today's reading = persistence); the forecast is $\max(0, \text{today} + \hat{\Delta})$, and the two forecasts are averaged with a fixed weight of 0.5. Same training rows and validation set as above.

- *Ridge half:* the 39 selected features, standardised (scaler fitted on training rows), α chosen on a grid: $\alpha = 316$ (RMSE 29.033 / MAE 16.534 vs 29.078 / 16.572 for Ridge on the level). The penalty now shrinks towards "no change" instead of towards the training mean.
- *Extra-trees half:* 5 ensembles × 60 trees, each ensemble on a random 40 % of the rows, $\frac{1}{3}$ of the features per split. Chosen on validation over feature set (selected groups / all but seasonal / all 8 groups) and leaf size (10 / 25 / 50): all 57 features, leaf 10 (blend RMSE 28.773; the other eight settings give 28.82–28.95).

Model (validation)	RMSE	MAE	RMSE Sep / Oct / Nov
Ridge, change target ($\alpha = 316$)	29.03	16.53	19.47 / 17.90 / 42.97
Extra-trees, change target (leaf 10)	29.29	16.26	19.36 / 18.23 / 43.41
Blend 0.5 / 0.5	28.77	16.13	19.26 / 17.78 / 42.58

- *Is the gain real?* The blend has the lower daily RMSE than the Part 2 Ridge on 55/91 days ($p = 0.029$), than the Part 1 Ridge on 60/91 ($p = 0.0016$) and than persistence on 73/91 ($p = 2.5 \times 10^{-9}$). Neighbouring days are correlated, so these p-values are optimistic.
- *Blend weight* (Ridge share 0.3 / 0.4 / 0.5 / 0.6 / 0.7): RMSE 28.89 / 28.82 / 28.77 / 28.76 / 28.78, MAE 16.11 / 16.11 / 16.13 / 16.17 / 16.23. The curve is flat, so 0.5 is kept and not tuned.
- *Bias:* $+0.09/-2.39/-4.60$ in Sep / Oct / Nov (Part 2 Ridge $+0.15/-1.81/-3.22$): it under-forecasts the November build-up slightly.

Answers to the Part 2 questions

Q: What did each feature group add? The station's own 24 h are essential (+7.0 RMSE when removed). The network level adds the most beyond that (+0.21): a single sensor is noisy, while the network mean is a low-noise estimate of the regional, weather-driven level. Neighbour means at 15/30/60 km (+0.11) act as a spatially smoothed persistence; trend (+0.06) and day of week (+0.04) add a little. In the tree half all 8 groups help (all features beat the selected groups: 28.77 vs 28.90).

Q: Which ideas did not help, and why? `history_days`, `seasonal` and `neigh_knn` in the Ridge, a log1p target (RMSE 47.24 without correction, 53.94 with smearing; the exponential back-transform explodes on high November rows, max

forecast 1,438 µg/m^3), and removing or down-weighting Jun–Aug targets (29.03 / 29.05: a change of < 0.05, too small to trust). The main reason is a *seasonal regime shift*: training (Jan→Aug) is dominated by pollution falling from winter into the monsoon, validation (Sep→Nov) by pollution rising into winter. Long-memory and climatology features teach a linear level model to revert to a multi-day or training mean, which in a rising season means forecasting too low; and the annual sin/cos values of Sep–Nov are never seen in training.

Q: Did a more flexible model beat Ridge? Not on its own and not on the level. The trees fit the training months much better (train RMSE 19.1–20.2 vs 26.9) but are worse on validation (30.5–33.2 vs 29.1): they over-fit the training season, and a tree trained on the level cannot predict outside the target range of its leaves, so it misses November's higher concentrations (Nov RMSE 45.0–50.7 vs 43.0). A plain average of level models (29.44) did not help either. Only after switching to the change target, where the target is nearly stationary across seasons, did the average with Ridge win (28.77): Ridge extrapolates smoothly, the trees add non-linear detail on ordinary days (lower MAE), and their errors are only partly correlated.

Q: What information would help most? A weather forecast for the target day (wind, mixing height, inversions, rain), which drives sudden changes such as the clean-out of 29 Nov; then emission information (crop-residue fires, festival days such as Diwali on 31 Oct–1 Nov 2024) and a flag for which readings were gap-filled upstream.

4 Part 3 — Lead time

The target stays at 12:00 on d+1; the issue time moves to 11:00, 09:00, 06:00, 00:00 on d+1 or 12:00 on d. For every lead all features are rebuilt from the data up to the new issue time and every learned model is *refitted* on the same training target days (Part 1 Ridge: 242 days; Part 2 models: 236 days), with hyper-parameters unchanged (not re-tuned per lead).

Lead (issue time)	Persistence RMSE / MAE	Part 1 Ridge RMSE / MAE	Part 2 Ridge RMSE / MAE	**Final model** RMSE / MAE
1 h nowcast (11:00, d+1)	23.95 / 10.36	18.95 / 8.67	18.83 / 8.63	**18.69 / 7.88**
3 h (09:00, d+1)	51.43 / 28.28	28.37 / 16.55	**25.97 / 13.94**	27.09 / **13.89**
6 h (06:00, d+1)	66.82 / 41.28	33.72 / 22.80	**29.83 / 17.60**	31.14 / 17.81
12 h (00:00, d+1)	72.74 / 46.74	31.70 / 21.00	29.85 / 18.18	30.47 / **17.68**
24 h day-ahead (12:00, d)	32.75 / 17.47	29.58 / 18.16	29.08 / 16.57	**28.77 / 16.13**

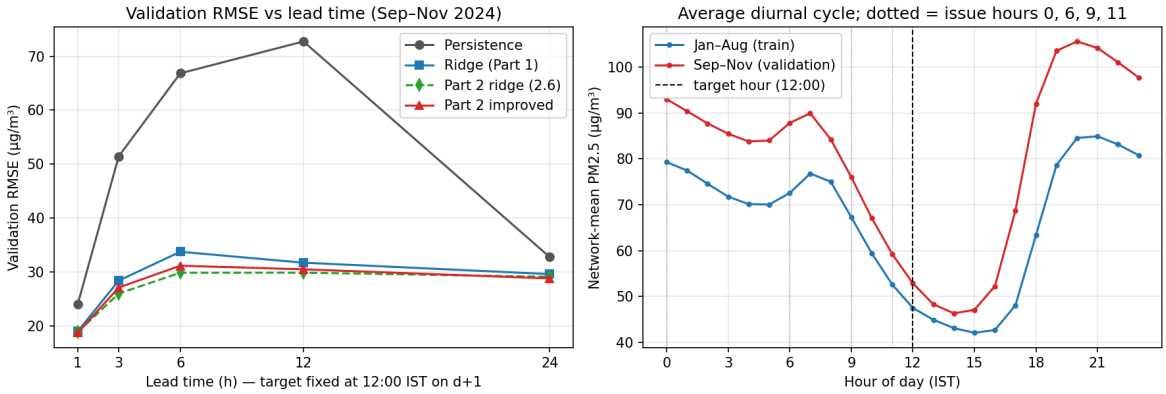


Figure 2: Left: validation RMSE against lead time (target fixed at 12:00 IST on d+1). Right: network-mean PM2.5 by hour of day for the training and validation months; dotted lines mark the issue hours 0, 6, 9, 11, dashed the target hour.

Q: Does persistence behave differently at 1 h, 24 h and in between? Yes, strongly and not monotonically: 23.95 at 1 h, then 51.43, 66.82 and 72.74 at 3, 6 and 12 h, and back down to 32.75 at 24 h. The cause is the daily cycle (right panel): PM2.5 is high at night and in the morning (shallow, stable boundary layer) and falls to its minimum in the early afternoon (around 14–15 h) when mixing is deepest; the noon target sits on this low. A reading from 00:00, 06:00 or 09:00 belongs to a much more polluted phase, so copying it is biased high. At 24 h the issue hour equals the target hour, the daily cycle cancels and only the day-to-day change remains. Even the 11:00 reading lies on the falling limb, so the 1 h error is not tiny.

Q: Do the learned models show the same behaviour? Only weakly. Being refitted per lead, they learn the hour-of-day offset and are 1.9–2.4× better than persistence at 3–12 h. They are best at 1 h (18.7–19.0) but are not monotone either: 6 h and 12 h are worse than 24 h for the Part 1 Ridge (33.72, 31.70 vs 29.58) and for the final model (31.14, 30.47 vs 28.77). How fast the night peak decays into the noon minimum depends on that day's mixing, which we do not observe, so a night reading is a noisier predictor of noon than yesterday's noon. The final model is the best at 1 h and 24 h only; at 3–12 h the Part 2 Ridge has the lower RMSE. The final model was tuned for 24 h, where y − today is a small day-to-day change; at 3–12 h that difference also contains the large, level-dependent drop from the morning peak to noon, which the tree half (limited to changes seen in Jan–Aug) under-predicts on polluted days while Ridge scales it linearly.

Q: Which lead for a public warning, and what does it cost? The 24 h lead issued at 12:00 on d: people need a day to plan. Against the 1 h nowcast it costs RMSE 28.77 vs 18.69 (+10.1 µg/m^3, +54 %) and MAE 16.13 vs 7.88 (about twice). It is not worse than the 6 h and 12 h leads (31.14, 30.47), so a longer warning costs nothing over a mid-range one; a nowcast is too late to act on but can update the warning (3 h: RMSE 26–27).

5 Part 4 — Forecast maps vs observed maps

Inverse-distance weighting (power 2, all 468 stations, assignment km coordinates) onto the 3,396-point 0.05° grid; the *same* weight matrix maps observations, the final model, persistence and the Part 1 Ridge, so map differences come only from the forecasts (nearest grid point to a station: 0.15 km). Days: **clean** = lowest observed station mean (15 Sep, 7.9 µg/m^3), **polluted** = highest (27 Nov, 175.0 µg/m^3), **worst** = highest station RMSE of our model (29 Nov).

Averages over the 91 validation days	Persistence	Part 1 Ridge	**Final model**
mean daily map RMSE (forecast map vs observed map)	15.37	15.42	**14.31**
mean daily pattern correlation (across grid cells)	0.580	0.605	**0.616**
mean daily station RMSE	28.57	26.00	**25.00**
pooled station RMSE	32.75	29.58	**28.77**

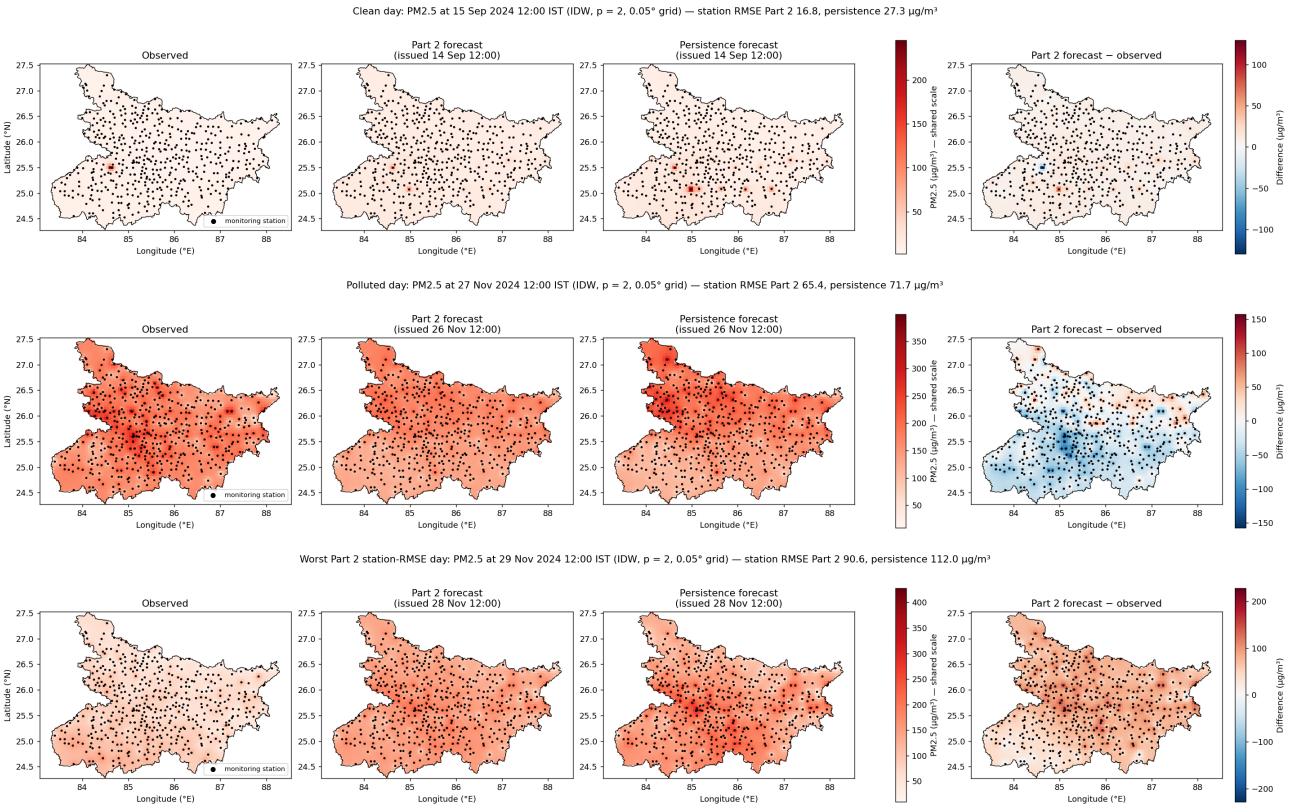


Figure 3: Observed, final-model forecast, persistence and forecast − observed for the clean (15 Sep), polluted (27 Nov) and worst (29 Nov) days; one shared colour scale per row for the first three panels. Station RMSE ours / persistence: 16.79 / 27.28, 65.39 / 71.67, 90.63 / 112.01; map RMSE ours 10.09, 36.98, 75.66; pattern correlation ours / persistence 0.62 / 0.41, 0.56 / 0.51, 0.39 / 0.49. On 29 Nov pollution dropped sharply overnight after the 27–28 Nov episode (station mean 76.6 µg/m^3), so the forecast is too high almost everywhere.

Q: Which RMSE is better, map or station? Which should be quoted? The map RMSE is much smaller (14.31 vs 25.00 for our model). That flatters the forecast: IDW averages many stations, so station-to-station errors cancel, densely sampled areas dominate, and the "truth" is itself an interpolation sharing the same smoothing. It also separates models poorly (persistence 15.37 vs Part 1 Ridge 15.42 on maps, although Ridge is much better at stations, 26.00 vs 28.57). The **station RMSE** is the number to quote: it compares with real measurements, as in Parts 1–3 and the hidden test.

Q: In what sense is the observed map not ground truth? It is built from 468 point sensors: between stations it is a weighted average, so it can never exceed the highest or fall below the lowest station value, it ignores real gradients (towns, roads, rivers, wind) and local sources, and its quality depends on station density. The station values themselves contain sensor noise and upstream gap-filling.

Q: Good pattern correlation but poor RMSE — which matters for a warning? Correlation ignores a uniform offset or scaling: a forecast that gets the spatial pattern right but is, say, 60 µg/m^3 too low everywhere (as when pollution jumps overnight) has correlation near 1 and a large RMSE. For a public warning the **level** matters more, because warnings are

issued when a threshold is crossed; pattern correlation is useful only for deciding *where* to target it.

6 Part 5 — Hidden December test

- `forecast.py` is the final model refitted with the same settings on all targets up to 30 Nov (327 days, 9 Jan–30 Nov, 153,036 rows): Ridge half ($\alpha = 316$, 39 features) and Extra-trees half (5×60 trees, leaf 10, 57 features), blend weight 0.5. The station means of the seasonal group are recomputed over the refit rows. Everything is stored in `model.pkl` (67.6 MB) and loaded once at import; `forecast()` only builds features from history up to issue time (the shared `pm25_features.py`) and calls predict — no refitting.
- *Tests:* inside the notebook `forecast.py` reproduces the in-notebook predictions of the refitted model and gives identical output when rows after the issue time are appended (leakage check); slowest call 0.06 s. `check_submission.py` passes (median 0.08 s per call, limit 10 s). Its November RMSE (36.76 vs 47.81 for persistence) is *in-sample*, since November is now training data, and is not a test score.
- *Expectation:* the model beat persistence in every validation month, so it should beat it in December, but this is not certain: only 30 days are scored, December is the most polluted month, and one sudden clean-out day like 29 Nov can dominate RMSE.

7 Limitations

Validation has only 91 days, dominated by November, and it was used to choose groups, hyper-parameters and the model, so the validation numbers are somewhat optimistic and differences of a few tenths (the final model gains 0.31 RMSE over the Part 2 Ridge) should not be over-interpreted. Training months show pollution falling, validation and December rising. The final model was tuned for the 24 h lead only. Bihar gaps were filled upstream and are not marked. The Part 0 calibration uses only `ok` readings and does not separate spikes; the 12-character Kanpur sensor appears to have a timestamp offset.

8 Use of AI tools

We used Claude (Anthropic) as an assistant throughout the project:

- *Writing and debugging code* for Parts 1–5: the shared feature builder `pm25_features.py`, group selection and ablation, model comparisons, the lead-time study, the IDW maps, `forecast.py`, and merging our improved Part 2 model into the final notebook.
- *Understanding concepts:* e.g. why a squared-error model can lose on MAE, regression dilution in the calibration check, the diurnal cycle behind the persistence curve, and map vs station RMSE.
- *Optimisation ideas:* suggestions for feature groups, for predicting the change from today's reading, and for blending Ridge with Extra-trees.
- *Plots and the report:* making figures, and drafting and formatting this report from our results.

The group checked every result (reproduction of the Part 1 reference numbers, leakage and `forecast.py` consistency checks, `check_submission.py`, re-runs from a fresh kernel), reviewed the code and the text, and all members understand the submitted work.